

Harouna Thiam

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EDUCATION

Duke University

Durham, NC

B.S. in Computer Science + Artificial Intelligence & Machine Learning

Expected May 2028

Relevant Coursework: DSA, Computer Systems, Discrete Mathematics, Probability, Distributed Systems, Linear Algebra

Awards: HackDuke '25 Winner, U.S. Congressional App Challenge Winner, **Amazon** Future Engineer Scholarship (1/400 selected), **LinkedIn** Possibilities Scholarship (1/25 selected), **Jane Street** UNBOXED Scholarship (1/38 selected), **Walmart GT** Summit Fellow (<1.5% selected), **MLT** Career Prep Fellow, **Goldman Sachs** Engineering Summit Fellow

SKILLS

Languages: Python, Java, React, SQL, Kotlin, TypeScript/JavaScript, R, Swift, Go, HTML/CSS

Infrastructure: AWS (Lambda, S3, Bedrock, SNS, Step Func.), Docker, git, CI/CD, REST, Kubernetes, JUnit, MySQL, kind

AI/ML/Data: LangChain, Weaviate, Pandas, Prompt Engineering, NumPy, Tensorflow, Codex, Claude Code, Cursor

EXPERIENCE

Duke University • Liberata Research Group • Incoming Machine Learning Intern

Aug. 2026 – Apr. 2027

Amazon Web Services • Elastic Kubernetes Services (EKS) • Incoming Software Development Intern

May. 2026 – Aug. 2026

Amazon • Devices (Tablet Launcher) • Software Development Intern

May. 2025 – Aug. 2025

- Built and owned serverless cloud workflow using Java and AWS to automate data augmentation and validation with LLMs
- Reduced JSON processing time by **87%** by using a distributed parallel map state with AWS Bedrock and AWS Step Functions
- Designed a callback token mechanism with TypeScript SNS listeners and Lambda gateways to synchronize validation responses
- Implemented an Utterance Validation Framework with handoffs to S3 between workflow states to validate Alexa text-to-speech
- Ensured code quality with automated testing suites (unit, integration, load) with JUnit, integrated into a CI/CD pipeline

Duke Fuqua School of Business • Software Engineer & Course Assistant

Apr. 2025 – Mar. 2026

- Built and deployed a course submission autograder using Python, Pandas, Openpyxl. Decreased grading time for TAs by **42%**
- Integrated generative AI prompts across **150+** pages of textbooks to create AI-assisted instructions for graduate coursework
- Facilitated the integration of a new Learning Management System (LMS) for AI-Enhanced graduate student learning
- Overseeing TA team for a course of **400+** students, delegated assignment grading for TAs

ZSuite Technologies • AI/ML Software Engineer Intern

Jun. 2023 – Aug. 2023

- Developed LLM chatbot tool with Python and Langchain, resulting in increased efficiency of company representative response
- Curated relevant context to an LLM from CSV files through a vector store with Weaviate and RAG, improving query accuracy
- Containerized code and workspace with Docker and Jupyter Notebooks to create a scalable and modular codebase

PROJECTS

Reaper: Kubernetes Controller for Gang Protection and Horizontal Autoscaling in Distributed ML Training

May. 2026

- Built a Kubernetes controller in **Go** automating GPU node lifecycles, automating **cluster** state changes with **reconciler** for EKS
- Designed an anti-disruption **policy layer** to automatically protect **GPU node gang** schedules from deprovisionment on **Karpenter**
- Validated end-to-end against a multi-node **kind** cluster with simulated GPU node pools and distributed training job manifests
- Designed first-fit **bin-packing** logic to detect consolidation candidates, flagging underutilized GPU nodes for Karpenter rightsizing
- Exposed **Prometheus** metrics for reclamation and eviction prevention rates; added controller **leader election** via Kubernetes **Lease**

Duke University 'Code for Good' Hackathon • 1st Place App in Sustainability

Feb. 2025

- Developed a full mobile application in a team setting at HackDuke, utilizing flutter to create an interactive daily challenge platform
- Utilized Gemini API to dynamically generate engaging personal challenges and gamified progress for users with LLMs
- Used game design and UI/UX principles to secure **150+** downloads with engaging storytelling and progress tracking

United States Congressional App Challenge • 1st Place App - Florida

Nov. 2023

- Integrated image classification models using TensorFlow to detect invasive species and diagnose plant conditions (mobile)
- Utilized Kotlin coroutines and XML to generate dynamic plant descriptions with OpenAI and asynchronous data flows
- Optimized mobile performance by using TensorFlow Lite for on-device inferences, reducing latency and ensuring offline access

LEADERSHIP ACTIVITIES & PROFESSIONAL DEVELOPMENT

Alliance for Innovative and Interdisciplinary Computing Education (AiiCE) • National Student Advisory Board Member

Cornell Tech: Break Through Tech • AI Program Fellow

LinkedIn • Possibilities [in] Technology Cohort Member

ColorStack@Duke & National Society of Black Engineers • Member